

# Nosherwan Babar

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## PROFESSIONAL SUMMARY

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MS Data Science candidate at Drexel University (GPA 3.75, graduating July 2026) with production experience building NLP and LLM-powered applications, model-backed APIs, and MLOps infrastructure. Completed a 6-month ML Engineer Co-op at URBN on GCP: built and deployed model APIs serving real-time predictions, implemented monitoring and automated retraining workflows, conducted bias analysis to support responsible model deployment, and maintained CI/CD pipelines with automated test coverage. Built CyberSentinel as a solo project: a production NLP and LLM platform with RAG pipelines, multi-agent orchestration, and end-to-end observability via LangChain tracing deployed on Cloud Run. Strong Python developer with hands-on NLP, GenAI, PyTorch, and cloud infrastructure experience across GCP and AWS.

## CORE COMPETENCIES

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NLP and LLM Application Development | ML Model Development and Deployment | Production API Development | MLOps and Model Monitoring | Responsible AI and Bias Analysis | Python and PyTorch | CI/CD and Automated Testing | Cloud Infrastructure (GCP, AWS)

## WORK EXPERIENCE

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### Machine Learning Engineer Co-op

URBN (Urban Outfitters Inc.)

Philadelphia, PA

September 2025 - March 2026

- Developed and deployed model-backed REST APIs using Python and FastAPI serving real-time ML predictions to internal product dashboards; integrated outputs into downstream workflows with observability logging, error handling, and performance monitoring.
- Built MLOps infrastructure on GCP: drift detection systems monitoring production model health and triggering automated Airflow retraining workflows when performance thresholds were breached; instrumented models with metrics and alerting to catch regressions early.
- Implemented responsible AI practices: conducted bias analysis at the class level and documented model assumptions and limitations before deployment; maintained evaluation suites and reproducible experiment pipelines throughout the model lifecycle.
- Designed and maintained distributed ETL pipelines (BigQuery, DBT, Airflow) processing large-scale transactional datasets with automated data validation; built and deployed predictive ML and statistical regression models with rigorous feature engineering.
- Implemented CI/CD pipelines with automated test coverage across GCP infrastructure ensuring reproducible workflows; wrote clear documentation and runbooks for operational support; participated in code reviews with cross-functional teams.

### Junior Software Engineer

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Lahore, Pakistan

August 2023 - August 2024

- Built and maintained backend microservices and REST/gRPC APIs using Python and SQL; diagnosed and resolved query bottlenecks through SQL optimization and caching, improving API latency by 15% across high-traffic endpoints.
- Configured AWS S3 for scalable media storage with automated backup pipelines; optimized complex SQL queries against relational databases; contributed to 4 full-stack product releases in Agile sprints with code reviews and QA collaboration.

## PROJECTS

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## CyberSentinel - Production NLP and LLM Platform

Personal Project

- Designed and built a production NLP and LLM application from scratch using Python, LangGraph, LangChain, and FastAPI: RAG pipeline over live CVE databases (ChromaDB with Gemini embeddings) for contextual, real-time NLP-driven retrieval and response generation.
- Implemented responsible AI observability via LangChain tracing to monitor agent behavior, surface failure patterns, and track LLM reliability; automated nightly Airflow DAG for data ingestion and index updates.
- Deployed on Google Cloud Run with GitHub Actions CI/CD and 332 tests passing; integrated prompt hygiene and content safety controls throughout the pipeline. Technologies: Python, LangGraph, LangChain, FastAPI, ChromaDB, Airflow, GCP, Docker.

## SignBridge - Real-Time ASL Translation System

Capstone Project (Team)

- Built the data pipeline, LLM integration layer, and web UI for a real-time ASL-to-English AI application: LangChain + Gemini 1.5 Flash via GCP Vertex AI for context-aware sentence generation from recognized sign sequences.
- Designed and deployed the full application on GCP Cloud Run with GCS for model and dataset storage; built Streamlit + streamlit-webrtc interface for live camera-based sign recognition. Technologies: Python, MobileNetV2, LSTM, LangChain, Gemini, GCP Vertex AI, Cloud Run, Docker.

## Spending and Churn Analysis

Academic Project

- Built an end-to-end ML pipeline in Python modeling customer behavioral patterns from multi-source transactional data using XGBoost, PyTorch, and Scikit-learn; conducted bias analysis at class level to support responsible model selection.
- Applied PCA and clustering for exploratory analysis; evaluated models with confidence intervals and per-class performance breakdowns; translated findings into recommendations with clear visualizations.

## EDUCATION

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### Master of Science in Data Science

Drexel University

Philadelphia, PA

Graduating July 2026

GPA: 3.75/4.00 | Courses: NLP with Deep Learning, Applied Machine Learning, Quantitative Data Analysis, Recommender Systems

### Bachelor of Science in Computer Science

University of Central Punjab

Lahore, Pakistan

July 2023

GPA: 3.41/4.00 | Courses: Data Structures and Algorithms, Machine Learning, Statistics and Probability, Distributed Systems

## SKILLS

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NLP and GenAI: NLP (tokenization, entity extraction, text classification), LangGraph, LangChain, RAG (ChromaDB), Prompt Engineering, LLM APIs, Gemini, OpenAI-compatible APIs

ML and Deep Learning: PyTorch, TensorFlow, Scikit-learn, XGBoost, LightGBM, Transformers, CNNs, MobileNetV2, LSTM, Regression, Classification, Transfer Learning, A/B Testing

MLOps and APIs: Airflow, DBT, Drift Detection, Model Monitoring, CI/CD (GitHub Actions), Docker, FastAPI, REST/gRPC APIs, Automated Testing

Cloud and Data Engineering: GCP (Cloud Run, BigQuery, Vertex AI), AWS S3, PySpark, SQL, ETL/ELT, Parquet, Distributed Computing

Programming and Tools: Python, C++, SQL, JavaScript, Git, Pandas, NumPy, Jupyter, Matplotlib, Seaborn